

Utkarsh Kumar

Machine Learning & Software Engineer | LLMs | RAG | AI Agents | Computer Vision
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Summary

Computer Science student with hands-on experience building LLM applications, RAG pipelines, AI agents, computer vision systems, and scalable backend services. Developed production-oriented AI solutions for candidate discovery, intelligent automation, anomaly detection, and real-time inference using Python, FastAPI, PostgreSQL, React, and PyTorch.

Education

Manipal Institute of Technology, Bengaluru, India	GPA: 9.3
B.Tech in Computer Science (Data Science)	2024 – 2028
Indian Institute of Technology, Guwahati	GPA: 8.3
B.Sc in Artificial Intelligence and Data Science	2024 – 2028

Technical Skills

Programming Languages: Python, C++, Java, JavaScript, TypeScript, SQL

Machine Learning / AI: LLMs, Retrieval-Augmented Generation (RAG), AI Agents, Multi-Agent Systems, LangChain, OpenAI API, Hugging Face, PyTorch, Scikit-Learn, Computer Vision, OpenCV, FAISS, PyTorch Geometric, NetworkX

Backend / Frameworks: FastAPI, Flask, Node.js, React.js, React Native, Next.js, Vite, TanStack Query, GraphQL, WebSockets, SQLAlchemy, Uvicorn, uvloop, Expo

Databases / Cloud / DevOps: PostgreSQL, MongoDB, Redis, Azure AI Search, AWS (EC2, S3), Google Cloud Platform, Supabase, Vercel, Docker, Git, GitHub

Data / Visualization: Pandas, NumPy, D3.js, Chart.js, Spatial Analytics

Experience

AI / Software Engineering Intern, Provaantech	Apr 2026 – May 2026
GitHub Demo	

- Engineered an AI Candidate Discovery Engine that scans 110M+ resumes and returns the top 10 matches in under 10 seconds.
- Built hybrid retrieval using OpenAI embeddings, Azure AI Search, BM25 ranking, and semantic search for precise candidate matching.
- Added GPT-4o-mini reasoning to generate automated fit explanations for shortlisted candidates.

Software Engineering Intern, AVSS Tech	Dec 2025 – Jan 2026
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- Built a real-time face recognition pipeline with 92%+ accuracy.
- Reduced inference latency by 35% using batching and frame optimization.

Researcher, IEEE GRSS, Manipal Institute of Technology	Jan 2025
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- Developed a two-stage hurdle stacking ensemble across 210 weather stations using 15 years of NASA data.
- Achieved $R^2=0.581$, $MAE=2.17$ mm, and Skill Score=0.510.
- Built leakage-free ML pipelines using chronological split and out-of-fold meta learning.

Selected Projects

CrowdWisdom.AI – Multi-Agent LLM System	GitHub Live Demo
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- Built a multi-agent orchestration system for automated workflows and task execution.
- Improved scheduling efficiency by 40% using intelligent agent routing.
- Integrated ElevenLabs Voice AI for real-time conversational interaction.

Gender Classification App	Live Demo
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- Built a real-time classification application with 90%+ prediction accuracy.

Awards & Achievements

AgniSense – Comprehensive Predictive & Operational Command Framework	GitHub
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- Top Winner, National-Level Hackathon (75,000 Prize).
- Built an enterprise-grade fire-risk prediction platform integrating 5 municipal datasets into a unified feature store.
- Developed a FastAPI + Uvicorn ML backend that processes 1.7+ GB telemetry data to serve real-time Khatra Score predictions.

SmurfPakad – Blockchain Fraud Detection via Graph Neural Networks	GitHub Demo
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- Top 3, State-Level Hackathon (20,000 Prize).
- Built a GNN-based fraud detection system using GraphSAGE to identify blockchain money laundering patterns.
- Delivered a full-stack solution with FastAPI, PostgreSQL, Docker, and D3.js visualizations.